VIAM

Robot Hackathon: Edge of Intelligence

November 13 & 14, 2026

ROBOT HACKATHON: EDGE OF INTELLIGENCE

Schedule

Day 1

- **9:00 AM:** Check-in and breakfast
- **9:30 AM:** Welcome & Kickoff
- **10:00 AM:** Teams and hardware
- **10:30 AM:** Hacking begins
- **1:00 PM:** Lunch
- **6:00 PM:** Dinner
- **9:00 PM:** HACKING ENDS

Day 2

- **9:00 AM:** Doors open, breakfast
- **9:30 AM:** Hacking resumes
- **12:30 PM:** Lunch
- **3:30 PM:** Demos and judging
- **5:00 PM:** Awards
- **6:00 PM:** Event concludes

Three tracks

- **Track 1, Teach it by showing.** Record demonstrations with the leader arm, train ACT or fine-tune SmolVLA on them, and run the policy on your station.
- **Track 2, Big models, small box.** Run pi0.5 on the station's NUC’s integrated GPU and have it pack from language orders. Start from a baseline fine-tuned model.
- **Track 3, Spot it, sort it.** Cameras and an anomaly model find defective items, something decides what to do with each, and the arm sorts them.

Track 1 & 2: The packing task

A bin holds mixed items, an open shipping box sits next to it, and item positions are randomized. Each judged run is ten minutes. You earn points by tier:

- Pick one item and place it in the box
- Pack the three items on the order and ignore two items that are not
- Fit an awkward item: a deformable or oddly shaped item
- Close the lid
- Recover when a judge moves an item mid-run
- (Track 2 only) Follow a conditional spoken order, such as "pack everything except the fragile one"

A freestyle demo counts for part of your score. Show the judges something the tiers do not cover.

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Track 3: Spot it, sort it scoring

One five minute run on 6 judge items (good, defective, and at least one defect type you haven't seen), then a five minute design explanation. Points for the following.

- Decision: right destination decision per item, from your log.
- Execution: items that ended where your system decided to send them.
- Latency: timestamp to first arm motion, correct decisions only.
- Recovery: keeps working when a judge moves an item mid-run.
- Design explanation: what it does, why each choice, what failed, what you would change.

Log one line per item: number, decision, confidence, timestamp. If there's no log, no Decision or Latency points.

Your hardware

- UFactory xArm6 or 850 robot arm and its control box
- UFactory two-finger gripper
- Intel RealSense D435 mounted on the arm
- A webcam and a stand
- Intel compute: Arc iGPU, an ASUS NUC or an AAEON box

What’s already configured

Resource	What it is
`arm, gripper`	The xArm6 and its gripper
`cam, overview-cam`	The wrist camera and the scene camera
`leader-arm, leader-gripper`	The SO-101 leader
`teleop`	Maps the leader's end-effector pose onto the arm through the motion service
`capture-control, data_manager`	What to record and when; the recording and the sync
`home-pose, leader-home`	Saved poses the webapp's H key sends the arms to
`capture-webapp`	The capture control webapp, served from the station

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Before the arm moves

- The E-stop is the red button on the xArm controller. Pressing it stops the arm at once. Release it to move the arm again.
- Speed caps apply to every move. Teleop caps the arm at 250 mm/s and 180 deg/s; the arm's own moves run at 60 deg/s. A cap slows the arm. It does not stop it.



Teleop

- On start, the leader moves itself to match the arm. Hands off until it stops
- The arm follows the leader's end-effector pose through the motion service, scaled 2.5 times. A five-joint leader drives a six-joint arm
- Squeeze the leader's gripper and the arm's gripper closes
- If the arm does not follow: on the teleop service's DoCommand card, send `{"command": "status"}` and read `pipeline_error`

Recording an episode

Keep the session FPS at 10 and type the task exactly. Training reads the task tag as the instruction.

Key	What it is
Enter	Starts a session
Space	Starts the first episode; after that, ends the episode and starts the next
ESC	Stops the session
T	Toggles teleop. While recording, stops teleop and ends the episode
H	Homes every saver in the home set. Not while recording
1 to 9	Picks the task for the next episode

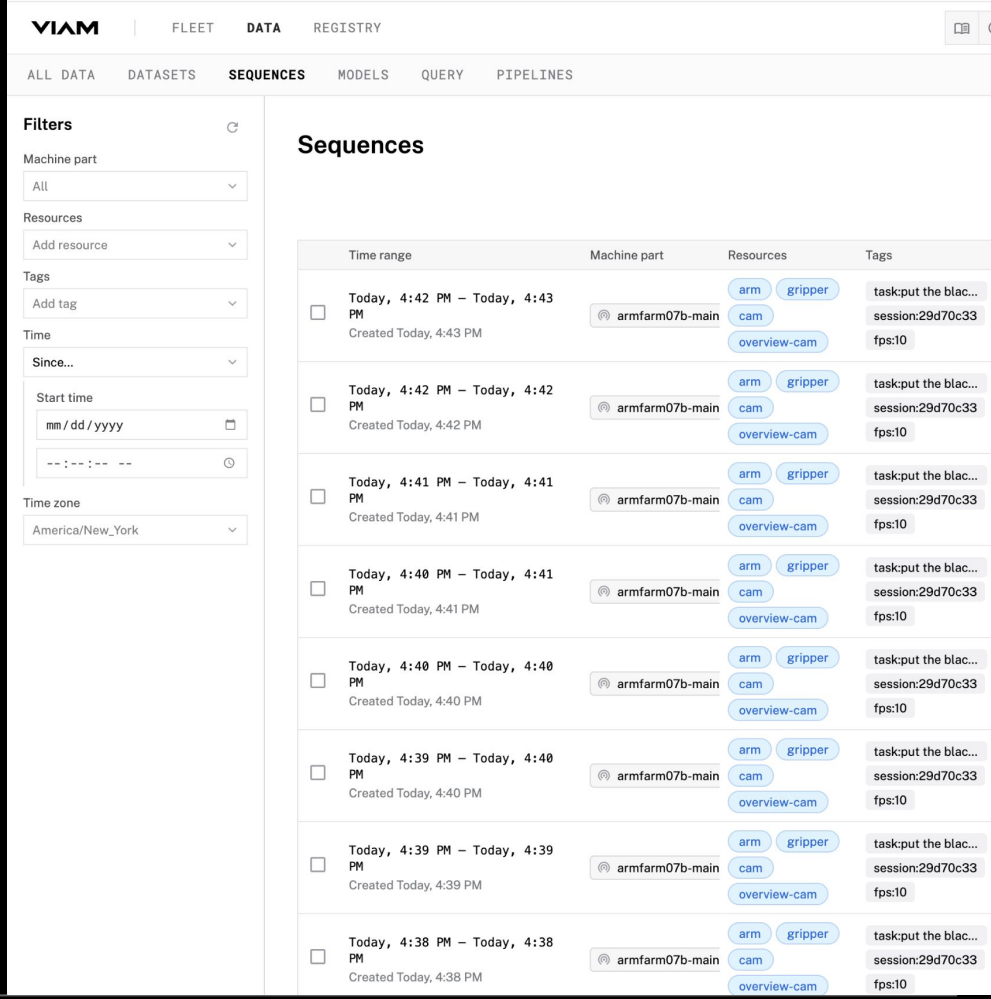
One episode is one sequence

- Tagged with `task:`, `session:` and `fps:`
- Holding four streams at the session's frame rate: the arm's joint positions, the gripper's state, the wrist camera's frames, the scene camera's frames
- Nothing is captured between episodes

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Where the episode appears

- The Viam app, DATA tab, SEQUENCES
- Each row: the time range, the machine part, the resources recorded, the tags
- Open a sequence, pick the wrist camera's images, View in data gallery
- Data syncs every 6 seconds; the first entries can take 30 to 60 seconds



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Join the hackathon Discord channel

- https://discord.gg/YnEMfZCRCf



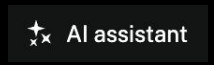
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Reference Sheet

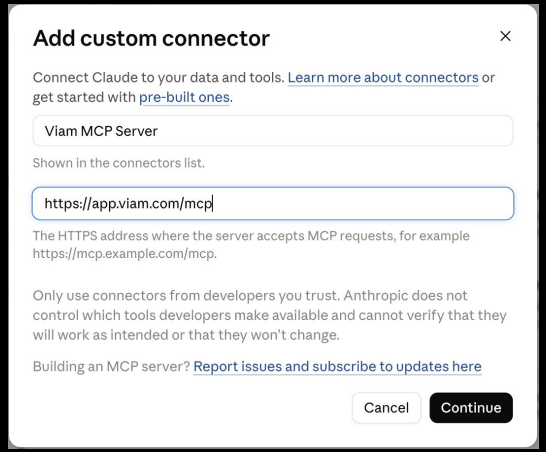
The sheet is in the event Discord: https://discord.gg/YnEMfZCRCf

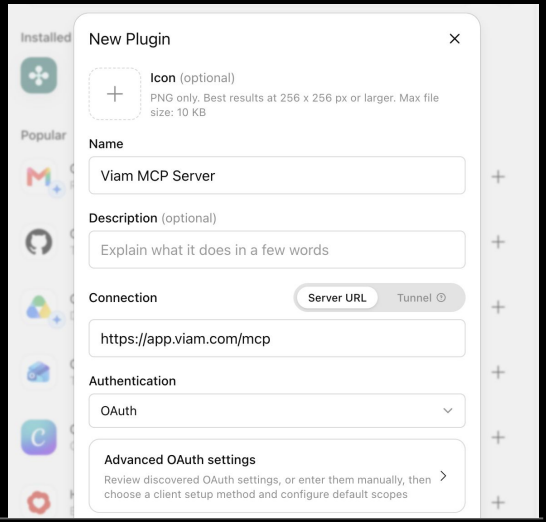
- Your station, before the arm moves, teleop (including what to do if the leader needs recalibrating) and recording: what you just watched
- One section per track: the starting point, the setup steps, the training command, the pre-warm list, and things to keep in mind
- Using AI and modules worth a look

Using AI

- The AI assistant in the Viam app 

- Your own AI agent: you can point it at Viam Docs or the agent-skills repository
- The Viam MCP server is available at https://app.viam.com/mcp
  - In Claude, go to Customize > Connectors > Add
  - In ChatGPT, go to Plugins > + Icon
    - You'll need to enable developer mode to be able to add custom plugins. You can do this by going to Settings > Security and login > Developer mode





Resources & Getting Unstuck

- Ask a Viam staff!
- Discord channel
- Viam docs
- Viam registry

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Share your build

- Excited about your build? Our video team's around if you want to talk about it or show it off.
- Tag us on socials @viamrobotics
- Share in the Robotics topic on Hackster.io
  https://www.hackster.io/robotics



Hackster.io

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Find your machine

- Go to https://app.viam.com
- Organization > location > machine
- Find your machine in the Arm Farm location of the hackathons organization

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Happy Hacking!


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